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Industrial sufficiency in manufacturing business models: development and validation of a business-model-level scoring model

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Manufacturing systems are major drivers of resource use and environmental impacts, and current trajectories indicate that efficiency and consistency strategies alone are unlikely to achieve the absolute reductions required to stay within planetary boundaries. Sufficiency, understood here as moderating absolute demand for material goods and intensifying the use of existing assets, is increasingly discussed as a third, complementary sustainability strategy alongside efficiency and consistency, but remains weakly operationalized at the business-model level, resulting in a low transparency on sufficiency. This article addresses the resulting measurement gap by developing and empirically validating a business-model-level scoring model for industrial sufficiency in manufacturing firms. Building on sufficiency literature and business-model frameworks, the study adopts a Design Science Research approach to derive three assessment dimensions: Value Proposition, Value Creation & Delivery, and Value Capture, operationalized through eight sufficiency-related parameters scored on a three-point ordinal scale. The model validated through five expert interviews with practitioners from manufacturing companies using a mixed-methods validation design that combines standardized ratings and qualitative comments. The results show that the model can be applied as a lean self-assessment tool, discriminates between different maturity levels across parameters, and, in the participating firms' self-assessments, highlights comparatively low maturity levels in parameters such as sufficiency in supply chain and logistics, pricing strategies for promoting sufficiency, and sufficiency-oriented value capture. The study concludes that the scoring model fills a gap between largely qualitative sufficiency frameworks and outcome-oriented assessment tools such as LCA or ESG metrics. For practitioners, it offers a structured basis for self-assessment and strategic prioritization of sufficiency measures at business-model level; for researchers, it provides an operational instrument that can be further tested, calibrated and extended in future empirical work.

KEYWORDS
industrial sufficiency, sufficiency economy, sustainability assessment, sustainability strategies, sustainable manufacturing

1 Introduction

Global environmental crises and the transgression of planetary boundaries increasingly call into question whether prevailing production and consumption patterns can be reconciled with long-term ecological limits (Niessen and Bocken, 2021; Richardson et al., 2023). In manufacturing, decades of corporate sustainability initiatives have focused predominantly on efficiency, that is reducing resource and emission intensities per unit of output, and on consistency or circularity, that is, closing material loops and substituting harmful materials (Bjørn et al., 2017; Bocken and Short, 2016; Schmidt, 2008). While these approaches have produced important relative improvements, their ability to secure absolute reductions in resource use and emissions is fundamentally constrained by rebound effects and continued growth in production volumes (Bocken et al., 2022; Guzzo et al., 2023; Hellweg et al., 2023; Santarius, 2012).

In response, sufficiency has gained prominence as a third sustainability strategy, complementing efficiency and consistency. Sufficiency is concerned with moderating absolute demand for material goods, limiting the scale and pace of production, and making more of existing stocks and infrastructures rather than continually expanding them (Andert et al., 2025; Jungell-Michelsson and Heikkurinen, 2022; Niessen and Bocken, 2021; Spangenberg and Lorek, 2019). In corporate and especially industrial practice, however, sufficiency still plays only a marginal role as an explicit sustainability strategy, even though recent literature increasingly discusses it as a transformative principle that calls for rethinking value propositions, production systems and revenue models (Andert et al., 2025; Beyeler and Jaeger-Erben, 2022; Bocken et al., 2022; Niessen and Bocken, 2021; Schmidhäuser et al., 2024).

At the same time, the conceptual progress in sufficiency research contrasts with a lack of operational tools for measuring industrial sufficiency at the level of business models. Existing instruments such as life-cycle assessment (LCA), absolute environmental sustainability assessments, ESG metrics and corporate sustainability indices focus primarily on environmental and social outcomes of products, processes or organizations (André, 2024; Bjørn et al., 2020; Bocken et al., 2022; Diez-Cañamero et al., 2020; Svensson et al., 2016). They offer little guidance on whether the underlying business-model logic is structurally oriented toward sufficiency, for example by moderating sales volumes, extending product lifetimes or constraining product variety. Qualitative sufficiency frameworks, in turn, map principles and business practices but do not provide clearly defined scales or thresholds that would allow manufacturing firms to systematically assess their sufficiency orientation over time (André, 2024; Beyeler and Jaeger-Erben, 2022; Bocken et al., 2022; Schmidhäuser et al., 2024; Svensson et al., 2016).

For manufacturing companies, this creates a dual challenge. Normatively, sufficiency is promoted as a key lever for aligning industrial activity with planetary boundaries. Operationally, however, firms lack instruments to assess sufficiency-oriented practices at business-model level and to integrate them into internal steering, governance and communication (André, 2024; Bocken et al., 2022; Niessen and Bocken, 2021; Schmidhäuser et al., 2024). Addressing this gap requires a measurement approach that is sufficiently simple for self-assessment in practice yet grounded in the sufficiency and sustainability assessment literatures.

This article addresses the measurement gap by developing and empirically validating a business-model-level scoring model for industrial sufficiency in manufacturing firms. The study is guided by the research question:

How can industrial sufficiency in the business models of manufacturing companies be systematically measured?

To answer this question, the article adopts the business-model perspective of value proposition, value creation & delivery and value capture (Bocken et al., 2014; Kobayashi et al., 2025; Niessen and Bocken, 2021; Osterwalder and Pigneur, 2009) and translates sufficiency principles into eight sufficiency-related parameters operationalized on a three-point ordinal scale. The scoring model is treated as an artefact within a Design Science Research (DSR) approach (Gregor and Hevner, 2013; Mettler, 2011; Peffers et al., 2007; Venable et al., 2016), iteratively refined through literature analysis, conceptual design and empirical validation with manufacturing firms.

The remainder of the paper is structured as follows. Section 2 reviews the sufficiency and business-model literature, discusses rebound effects in production systems, and introduces business-model frameworks as the assessment lens. Section 3 describes the research design, model development and empirical validation. Section 4 presents the resulting scoring model and the findings from its application. Section 5 discusses theoretical and practical implications, the complementarity to established assessment frameworks, and limitations. Section 6 concludes and outlines avenues for future research.

2 Theoretical background

2.1 Sufficiency as part of the three sustainability strategies

Corporate sustainability discourse has traditionally revolved around two strategies: efficiency and consistency (Andert et al., 2025; Niessen and Bocken, 2021; Schmidhäuser et al., 2024). Efficiency aims to "do more with less" by reducing resource and emission intensities per unit of output, for example through energy-efficient technologies or lean production (Santarius et al., 2023; Speck et al., 2022). Consistency or circularity seeks to "close loops" by designing products and systems that rely on renewable resources, enable high recycling rates and minimize waste (Ayres and Ayres, 2002; Bocken et al., 2014; Brinken et al., 2022), compare Table 1. Both strategies are essential, yet they offer no guarantee that total environmental pressures will decline if overall production volumes continue to grow (Bocken and Short, 2016; Huber, 1999; Santarius, 2012; Wüst et al., 2022).

Rebound effects illustrate this tension. Efficiency gains can lower costs and thereby induce additional demand, such that absolute resource use and emissions do not decline as expected (Huber, 1999; Jungell-Michelsson and Heikkurinen, 2022; Niessen and Bocken, 2021; Santarius, 2012). Recent reviews emphasize that rebound mechanisms operate at multiple levels, including direct, indirect and macro-structural mechanisms, and can undermine sustainability goals even when relative efficiency and circularity improve (Bocken and Short, 2016; Guzzo et al., 2023; Lange et al., 2021).

Sufficiency is considered a third sustainability strategy that explicitly addresses these limitations by questioning the overall scale and pace of economic activity (Jungell-Michelsson and Heikkurinen, 2022; Sachs, 1993; Spangenberg and Lorek, 2019). From a conceptual perspective, sufficiency can be defined as "a strategy for moderating the scale of economic throughput and consumption to levels

TABLE 1 Comparison of the three sustainability strategies along key comparative dimensions.

Strategy	Approach	Industrial application	Limitations	Source
Efficiency	Do more with less	High	Rebound effects	Santarius et al. (2023)
Consistency	Close resource loops	Rising	Rebound effects, change of resource use	Brinken et al. (2022)
Sufficiency	Moderate activities within planetary boundaries	Low	Opposing today's business logic, acceptance	Niessen and Bocken (2021)

compatible with ecological and social limits, while maintaining a good life within planetary boundaries" (Spangenberg and Lorek, 2019; Niessen and Bocken, 2021; Jungell-Michelsson and Heikkurinen, 2022). In practice, sufficiency may involve restricting product variety, designing for long lifetimes and repair, moderating output growth expectations, and aligning corporate governance with the idea of "enough" rather than "more" (Beyeler and Jaeger-Erben, 2022; Bocken et al., 2022; Niessen and Bocken, 2021). While efficiency seeks to reduce resource and emission intensities per unit of output and consistency aims to close material loops and substitute harmful inputs, sufficiency addresses the overall scale of production and consumption, targeting absolute reductions that neither efficiency nor consistency can reliably deliver in the presence of rebound effects and continued output growth.

In industrial contexts, sufficiency must therefore be understood not only as an efficiency-enhancing design principle at product level, but as a strategic orientation that shapes business models, supply chains and organizational decision-making (Niessen and Bocken, 2021; Bocken et al., 2022; Schmidhäuser et al., 2024).

2.2 Sufficiency in business practice and business models

The academic debate on sufficiency has expanded rapidly in recent years. Jungell-Michelsson and Heikkurinen provide a systematic literature review that situates sufficiency within broader sustainability discourses and highlights its underdeveloped operationalisation, particularly in business contexts (Jungell-Michelsson and Heikkurinen, 2022). Niessen and Bocken propose the "business for sufficiency framework," which conceptualizes how companies can contribute to sufficiency by adjusting their value propositions, production systems and revenue models. They emphasize practices such as offering durable, repairable products, reducing product variety, and decoupling revenue from pure sales volume (Niessen and Bocken, 2021).

Beyeler and Jaeger-Erben empirically investigate sufficiency characteristics in business practices and identify recurring patterns across sectors, including deliberate limitation of offerings, degrowth-oriented strategies and new forms of provider–user relationships. Their analysis suggests that sufficiency is not limited to niche initiatives but can be integrated into mainstream business models, albeit often in tension with prevailing growth logics (Beyeler and Jaeger-Erben, 2022).

At a more systemic level, Gossen and Niessen compile conceptual and empirical contributions on "Sufficiency in Business" (Gossen and Niessen, 2024), framing sufficiency as a transformative potential for sustainability. Bocken et al. examine 150 companies to derive a "sufficiency-based circular economy" perspective, showing how sufficiency principles can be integrated into circular offerings, business models and value creation systems (Bocken et al., 2022). Another approach of Schmidhäuser et al. separates three sufficiency strategies (frugality, longevity, and specifity) in three distinct business determinants (product, production, and business model) (Schmidhäuser et al., 2024).

These contributions document a growing repertoire of sufficiency-oriented business practices and provide conceptual frameworks for understanding them. However, they are largely qualitative and descriptive. They offer patterns and archetypes but do not provide standardized indicators or scoring systems that would allow manufacturing firms to assess and compare their sufficiency orientation in a structured way (Bocken et al., 2022; Jungell-Michelsson and Heikkurinen, 2022; Schmidhäuser et al., 2024). This measurement gap is particularly pronounced at the level of business models, where the interplay of value proposition, value creation and value capture determines whether sufficiency is structurally embedded or only marginally implemented. Despite this growing body of work, sufficiency still occupies a niche position in corporate sustainability strategies, especially in manufacturing, where it rarely appears as an explicit, company-wide sustainability strategy (Bocken et al., 2022; Gossen and Niessen, 2024; Schmidhäuser et al., 2024).

2.3 Business model frameworks as assessment lens

Business models describe how organizations create, deliver and capture value (Bocken et al., 2014; Osterwalder and Pigneur, 2009; Richardson et al., 2023). A widely used conceptualisation distinguishes three core components: value proposition (what value is offered to whom), value creation & delivery (how the value is produced and delivered), and value capture (how revenues are generated and costs are structured) (Bocken et al., 2014; Osterwalder and Pigneur, 2009; Richardson et al., 2023). This triadic structure was selected as the assessment lens for the present study because it provides a parsimonious yet comprehensive abstraction of business-model logic that covers all core organizational domains relevant to sufficiency, and because it has been applied as an analytical framework in sufficiency research, enabling a literature-grounded mapping of sufficiency parameters to distinct business-model responsibilities (Bocken and Short, 2016; Bocken et al., 2022). This "magic triangle" is echoed across multiple sustainability-oriented business-model frameworks and provides a simple yet powerful structure for analyzing how sustainability goals are embedded in organizational value logic (Bocken et al., 2014; Gassmann et al., 2024).

For sufficiency, this triad is particularly useful because sufficiency can manifest differently across the three components (Bocken et al., 2014; Gossen and Niessen, 2024; Niessen and Bocken, 2021):

- In the value proposition, sufficiency relates to needs orientation, sufficiency communication and the deliberate limitation of supply that stimulate over-consumption (Beyeler and Jaeger-Erben, 2022; Bocken and Short, 2016; Niessen and Bocken, 2021).

- In value creation & delivery, it concerns product lifetime, repair and reuse services and Sufficiency in supply chain and logistics (Bocken and Short, 2016; Niessen and Bocken, 2021; Schmidhäuser et al., 2024).
- In value capture, this logic is operationalized through Pricing strategy to promote sufficiency and Sufficiency-oriented value skimming, reflected in pricing and revenue models that incentivize efficient use, make intensive use economically attractive, and avoid rewarding volume growth alone (Bocken and Short, 2016; Bocken et al., 2022; Schmidhäuser et al., 2024).

Using this triad as an assessment lens enables a business-model-level operationalization of sufficiency that can be linked to managerial responsibilities and governance structures. It also allows sufficiency-oriented practices to be captured in a way that complements, rather than duplicates, outcome-oriented environmental and social metrics.

2.4 Existing assessment frameworks and measurement gap

Established assessment frameworks such as life-cycle assessment (LCA), absolute environmental sustainability assessments, ESG metrics and Triple Bottom Line (TBL) approaches play an important role in evaluating corporate sustainability performance (Bjørn et al., 2020; Da Cunha et al., 2025; Hellweg et al., 2023; Nica et al., 2025). LCA and related methods assess environmental impacts along product life cycles and are increasingly extended toward absolute sustainability by relating impacts to planetary boundaries or other carrying capacities (Bjørn et al., 2017, 2020; Hellweg et al., 2023). ESG metrics and corporate sustainability indices aggregate environmental, social and governance indicators at company or portfolio level and are used for reporting and investment decisions (Clément et al., 2022; Da Cunha et al., 2025; OECD, 2025; Rau and Yu, 2024).

However, these instruments largely operate at product, process or organizational outcome level. They do not explicitly assess whether the underlying business-model logic is oriented toward sufficiency, for example whether companies deliberately limit product variety, align revenue models with intensive use, or moderate growth expectations (André, 2024; Bocken et al., 2022; Clément et al., 2022). At the same time, existing sufficiency frameworks for business tend to be qualitative, focusing on conceptual taxonomies and practice descriptions without providing clearly defined scoring rules that would support systematic self-assessment (André, 2024; Bocken et al., 2022; Gossen and Niessen, 2024).

Against this backdrop, there is a need for a pragmatic, business-model-oriented measurement instrument that translates sufficiency principles into a small set of clearly defined parameters and scoring rules. The next section describes the design and validation of such an instrument for manufacturing firms.

3 Methods

3.1 Research design: design science research and mixed methods

The study follows a Design Science Research (DSR) approach in which the sufficiency scoring model is conceived as an artefact that addresses a relevant problem, namely the lack of a business-model-level measurement approach for industrial sufficiency (Mettler, 2011; Peffers et al., 2007; Venable et al., 2016). DSR foregrounds iterative cycles of problem diagnosis, artefact design, evaluation and refinement, combining theoretical grounding with practical relevance.

Methodologically, the study uses a mixed-methods design that combines quantitative and qualitative elements (Creswell and Plano Clark, 2007; Guzzo et al., 2023; Patton, 2002). Quantitative parameters consist of numerical self-ratings (e.g., percentages) on the sufficiency parameters and validation criteria, collected on a three-point ordinal rating scale (coded 0–2). Qualitative data consist of experts' open comments on definitions, scales, thresholds and measurement approaches. Both strands are collected within a single expert-interview-based validation study and analyzed in an integrated manner to inform model refinement.

3.2 Development of the sufficiency scoring model

The development of the scoring model comprised three main steps:

Derivation of assessment dimensions and parameters. Based on sufficiency literature and business-model frameworks (Mettler, 2011; Niessen and Bocken, 2021; Kobayashi et al., 2025), the thesis derived three assessment dimensions aligned with the business-model triad: Value Proposition, Value Creation & Delivery, and Value Capture. Within these, eight sufficiency-related parameters were defined to capture key levers of industrial sufficiency in manufacturing, compared to Figure 1.

Operationalization through scoring scales and indicative thresholds. Each parameter was operationalized using a three-point ordinal scale (scores 0–2). The scale reflects increasing implementation and strategic anchoring of sufficiency practices. For parameters with a more quantitative character, such as product lifetime or repair and reuse services, indicative percentage- or indicator-based thresholds were proposed for each score, drawing on literature and pragmatic estimations (e.g., minimum lifetime extensions, service revenue shares). For more qualitative parameters, such as sufficiency-oriented communication, the levels were defined through qualitative descriptions of implementation degrees (e.g., absence of manipulative messaging), prominence of sufficiency in communication (Beyeler and Jaeger-Erben, 2022; Kobayashi et al., 2025; Niessen and Bocken, 2021).

Aggregation logic and scoring procedure. The scoring logic is deliberately kept simple to support self-assessment. Parameter scores (0–2) are aggregated into dimension scores on a 0–100 scale by summing the parameter scores within each dimension and normalizing. No single overall sufficiency index is calculated in the core model. Instead, the three-dimension scores provide a profile of sufficiency maturity across the main business-model components. This echoes calls in the sustainability assessment literature to represent sustainability performance in a multi-dimensional way and to avoid masking trade-offs between dimensions through excessive aggregation (Bocken et al., 2014; Gossen and Niessen, 2024). This paper emphasizes that the model is designed as a lean self-assessment instrument rather than a benchmarking tool.

Throughout these steps, design requirements were derived from literature on sufficiency, sustainability assessment and composite indicators, including transparency of scoring rules, a limited number of

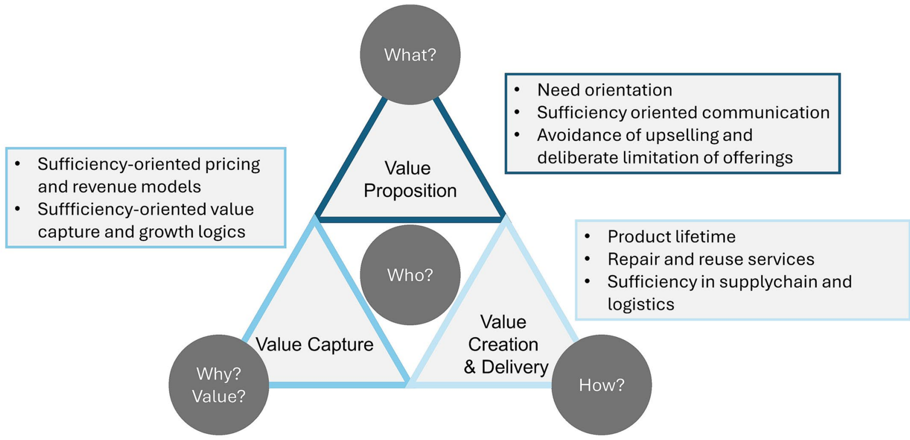

FIGURE 1
Conceptual allocation of sufficiency parameters to the three business model dimensions (based on the Magic Triangle by Gassmann et al.). Value Proposition: need orientation; sufficiency communication; supply limitation. Value Creation & Delivery: product lifetime; repair and reuse services; sufficiency in supply chain and logistics. Value Capture: Pricing strategy to promote sufficiency; sufficiency-oriented value capture and growth logic.

TABLE 2 Interview partners for the validation.

Industry	Company size	Role in the company	Relation to sustainability
Charging Infrastructure	Start-up (~15 employees)	Managing Director/Co-Founder (CFO)	Sustainability in investment, procurement, and operation of charging infrastructure (energy efficiency, durability, green electricity)
Packaging Machinery	Large Enterprise (~2,600 employees)	Innovation Manager (Technology and Business Model Innovation)	Machine and business model innovations (material/energy savings, modularity)
Mechanical Engineering	Small Enterprise (~30 employees)	Technology Officer (strategic technology responsibility, system architecture)	Responsibility for energy-efficient, renewably powered DAC technologies
Reusable Systems for Gastronomy	SME (~90 employees)	Product Lead (product development, material development)	Circular product development (durability, reparability, recyclability)
Bicycle Manufacturer	Large Enterprise (~900 employees)	Head of Sustainability (program management, KPI systems, supply chain)	Steering sustainability

parameters, and interpretability for practitioners (Diez-Cañamero et al., 2020; Hevner et al., 2024; OECD, 2025).

3.3 Empirical validation

3.3.1 Sample and recruitment

The empirical validation involved expert interviews with representatives from manufacturing firms. The target group comprised practitioners with responsibility for sustainability, strategy or business-model development in sectors such as mechanical engineering and industrial equipment. Five experts from different companies participated in the validation study. The selection is shown in Table 2. They were selected purposively to cover a range of company sizes and sufficiency maturity levels while ensuring that participants had sufficient familiarity with their firm's business model and sustainability initiatives. The sample was scoped in line with DSR practice for an initial design-evaluation cycle, which prioritizes conceptual clarity and practical relevance over statistical representativeness (Venable et al., 2016; Peffers et al., 2007). All participants held senior positions with direct responsibility for sustainability strategy or business-model development, including roles such as Chief Financial Officer, Head of Sustainability, and Technology Officer, ensuring the organizational authority and overview required to engage meaningfully with all assessment dimensions.

3.3.2 Data collection

The validation focused on assessing and refining the conceptual clarity, relevance and practical usability of the scoring model. To this end, an interview guide and a standardized validation template were developed. The guide and template were structured so that five quality criteria were assessed for each of the eight sufficiency parameters:

1. Comprehensibility of definitions and wording,
2. Perceived relevance for industrial sufficiency,
3. Measurement and practical applicability,
4. Appropriateness of scales and thresholds, and
5. Suggestions for improvement.

The interviews followed a two-step structure. First, participants were asked to provide an internal self-assessment of their company's

TABLE 3 Sufficiency scoring dimensions for manufacturing companies.

Sufficiency valuation criterion		
Valuation dimension: Value proposition		
Needs-orientation: Means fulfilling basic human needs within planetary boundaries while sales orientation aims to maximize sales and thus often counteracts sufficiency goals.	*Sufficiency communication*: Describes corporate communication that encourages conscious and resource-saving consumption. It shows how products can be used sensibly, long-lastingly and with the lowest possible consumption of resources.	*Supply limitation*: Describes a policy or structural measure to limit production and consumption to an ecologically acceptable level in order to reduce the planetary boundaries.
Valuation dimension: Value creation & delivery		
Product lifetime: Describes the actual or projected useful life of a product. Sufficiency oriented strategies aim to extend this duration in order to conserve resources, reduce environmental impact and avoid excessive consumption – without reducing the benefits for customers.	*Repair and reuse services*: Describe entrepreneurial measures for the targeted extension of product use. They reduce the need for new production, save resources and form central elements of sufficiency-oriented business models with a lower environmental impact.	*Sufficiency in supply chain and logistics*: Describes the deliberate and systematic limitation of production, transport and distribution activities to an ecologically sustainable and socially acceptable level, with the aim of avoiding or reducing resource consumption, emissions, overproduction and overdistribution – while at the same time ensuring that demand is met.
Valuation dimension: Value capture		
Pricing strategy to promote sufficiency: Describes the targeted use of economic incentives (e.g., through discounts or bonus systems) to reduce resource consumption through reuse, repair and use instead of ownership. The aim is to reduce the absolute volume of consumption while at the same time maintaining basic needs satisfaction.	*Sufficiency-oriented value capture and growth logic*: Refers to the generation of revenue through business models with a focus on absolute resource reduction, product life extension and needs-oriented use. The basis for evaluation is the percentage of these revenues in total sales, e.g., from repairs, rental models, upcycling or product service systems.	

current maturity for each parameter using the three-point scale (0–2), based on the parameter descriptions and, where available, indicative thresholds. Second, they evaluated the parameter and its scoring logic against the five validation criteria using the same three-point scale and open comment fields.

All closed validation elements were rated on the three-point scale (0–2) analogous to the scoring logic of the assessment model. Responses were converted into numerical codes for analysis and complemented with qualitative feedback in open comment fields. Data collection took place via online videoconferencing (MS Teams), with the interviewer sharing the validation template on screen and documenting ratings and comments during the sessions.

3.3.3 Data analysis

The quantitative and qualitative parts of the validation were analyzed in an integrated mixed-methods manner (Creswell and Plano Clark, 2007; Patton, 2002). For the quantitative part, descriptive statistics (e.g., mean values and ranges) were calculated for the self-assessment scores and for the five validation criteria across all parameters and interviews. This allowed an assessment of the model's discriminating power (variability of scores), perceived comprehensibility and relevance, and the adequacy of scales and thresholds.

The qualitative feedback was documented directly in the validation template and supplemented by short notes taken during the interviews. The written comments were consolidated after the data collection phase and obvious duplicates were removed. Where different experts raised similar points, these overlaps were noted.

The resulting comments clustered into four recurring themes:

i needs for clarification regarding parameter definitions and system boundaries,
ii suggestions to adjust scoring levels and indicative thresholds,
iii requests to specify evidence requirements and sub-indicators, and.
iv references to contextual conditions (e.g., sectoral characteristics, B2B vs. B2C markets).

This structured condensation served to link descriptive patterns in the quantitative results with concrete suggestions for change and to derive targeted impulses for iterative model refinement within the DSR cycle.

4 Results

4.1 Structure and logic of the scoring model

Table 3 presents the sufficiency scoring parameters for manufacturing companies. The model operationalizes the business-model "magic triangle" as conceptualized by Gassmann et al. by structuring the assessment along three business-model dimensions, specifically Value Proposition, Value Creation & Delivery, and Value Capture, and assigning eight sufficiency parameters to these dimensions.

In application, firms rate each parameter on the three-level ordinal scale (0–2) specified in Section 3.2, using the parameter descriptions and the guidance on measurement and evidence provided in Section 4.2. For the quantitative parameters, the scoring levels are supported by indicative threshold values (e.g., minimum lifetime extensions, shares of repaired products, or service revenue shares). For qualitative parameters, the levels are defined through concise descriptions of implementation depth and strategic anchoring.

Parameter scores are aggregated within each dimension and normalized to yield three-dimension scores on a 0–100 scale. In line with the model's purpose as a lean self-assessment instrument, results are reported as a three-part sufficiency profile rather than a single overall index. This enhances transparency regarding differences across business-model dimensions and supports the identification of targeted improvement areas.

4.2 Self-assessment results from validation with the companies

The self-assessments conducted as part of the validation indicate that the scoring model can discriminate between different maturity levels of sufficiency performance across the eight parameters.

Figure 2 summarizes the companies' self-assessment results across the eight scoring parameters. The blue bars indicate the mean parameter scores (0–2 scale) across companies, while the error bars represent the dispersion of ratings around the mean (e.g., standard deviation or standard error), thereby illustrating the degree of consensus and uncertainty in the self-assessments.

Across cases, relatively higher scores were observed for parameters linked to product longevity and repair and reuse services where firms had already implemented maintenance contracts, refurbishment programs or spare-parts offerings (compare to Figure 2). In contrast, lower scores were more frequent for the parameters "sufficiency in supply chain and logistics" and "sufficiency-oriented value capture," where practices such as sufficiency-oriented supplier criteria, low-carbon logistics strategies, or revenue models that reward intensive rather than extensive use are still emerging.

The dimension scores highlight typical patterns in how sufficiency is embedded in business models. In several cases, Value Proposition and elements of Value Creation & Delivery show higher maturity, reflecting sufficiency-oriented product features and service offerings, while Value Capture lags behind in terms of pricing and revenue models that align with sufficiency goals. This pattern is consistent with the literature, which notes that sufficiency-oriented revenue models and growth logics are particularly challenging to implement within existing market structures (Beyeler and Jaeger-Erben, 2022; Bocken et al., 2022; Gossen and Niessen, 2024).

Overall, sufficiency has so far been operationalized only to a limited extent as a corporate sustainability strategy, and the sufficiency performance of manufacturing firms remains at a generally low level. The analysis of the results shows that the companies' self-assessments were comparatively positive, indicating a tendency toward lenient evaluations. Therefore, the results should be interpreted with caution.

4.3 Validation results on quality criteria

The validation ratings for the five quality criteria provide further insight into the model's strengths and areas for improvement. Overall, experts rated the comprehensibility and perceived relevance of the parameters as high, with most scores at 1 or 2 on the three-point scale. This indicates that the chosen parameters resonate with practitioners' understanding of industrial sufficiency and can be communicated clearly.

Ratings for measurement and practical applicability as well as for scales and thresholds were more heterogeneous. For parameters with clearly definable quantitative indicators, such as product lifetime or

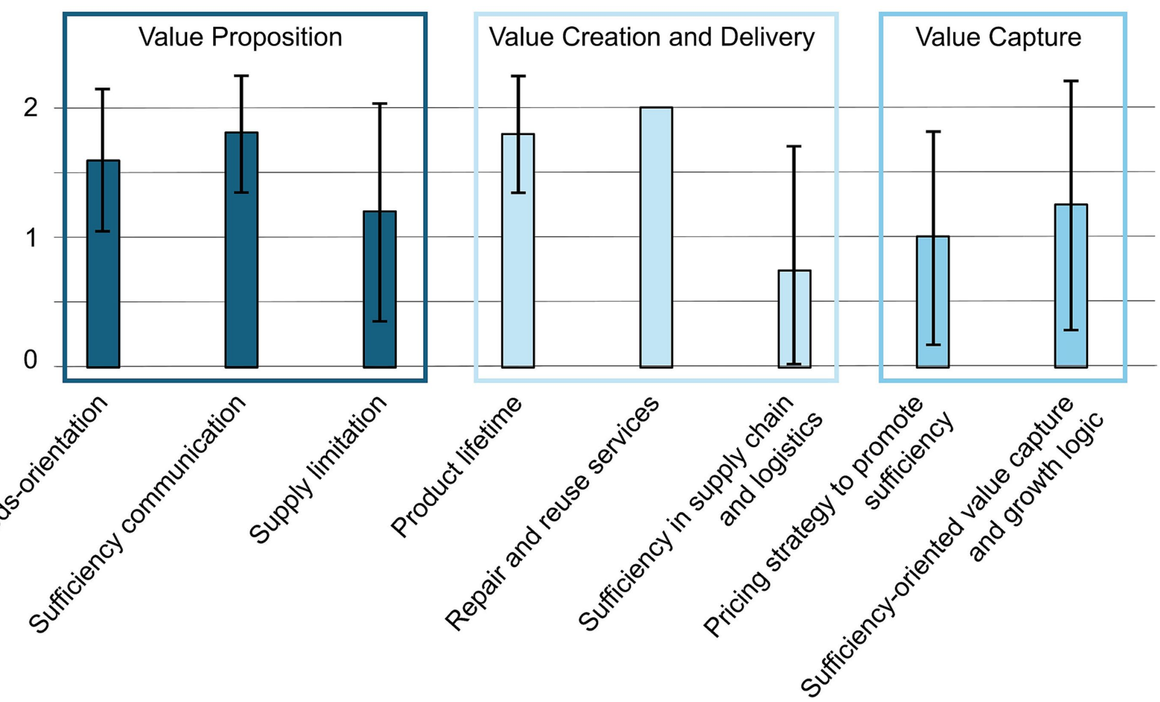

FIGURE 2
Sufficiency scores acquired in the validation process.

repair and reuse services, experts found the suggested thresholds and measurement approaches helpful but sometimes requested sector-specific benchmarks or more detailed guidance. To achieve a more detailed level of precision in the evaluation, this implies that additional documents with a lower degree of subjectivity (e.g., reports of revenue in certain fields, KPIs, CSR reports) could be incorporated into the assessment. It should be ensured, however, that the required effort remains proportionate to the level of accuracy provided by the three-point ordinal scale. For more qualitative parameters, such as sufficiency-oriented communication or value capture, they emphasized the need for clearer evidence examples and, in some cases, additional intermediate levels to capture partial implementation.

The qualitative feedback underscores these findings. Experts proposed clarifications of system boundaries, for example whether sufficiency in supply chain and logistics should focus on direct tier-1 suppliers or consider broader upstream effects. They also highlighted the importance of linking parameter definitions to existing internal KPIs and management systems to reduce additional data-collection burdens.

4.4 Summary of the model's further development opportunities

In summary, the validation suggests that the basic structure and logic of the sufficiency scoring model are robust, while indicating specific aspects that could be further developed in subsequent model iterations:

- The three-dimension structure and the selection of eight parameters can and should be retained.
- Definitions could be refined in future iterations to reduce ambiguity, particularly with regard to system boundaries and examples.
- For some parameters, indicative thresholds could be adjusted or formulated more cautiously to account for sectoral differences and data availability.
- Examples of evidence could be expanded to more clearly illustrate how self-assessments can be substantiated with appropriate documentation and indicators. This could be reached through the addition of further documents and KPIs.
- Parameters such as "sufficiency in supply chain and logistics" and "sufficiency-oriented value capture" should be explicitly marked as areas where further operationalization and empirical testing are required.

Overall, the results support the use of the model as a lean, practice-oriented self-assessment tool for manufacturing firms, while at the same time highlighting clear entry points for its gradual further development.

5 Discussion

5.1 Theoretical contribution: operationalizing sufficiency at business-model level

The scoring model contributes to sufficiency research by operationalizing industrial sufficiency at the level of business-model logic. Building on conceptual work that emphasizes sufficiency as a complementary sustainability strategy (Bocken et al., 2022; Jungell-Michelsson and Heikkurinen, 2022; Spangenberg and Lorek, 2019), the model translates principles such as need orientation, moderation of demand and intensive use into concrete assessment parameters aligned with the business-model triad of Value Proposition, Value Creation & Delivery, and Value Capture (Bocken et al., 2014; Osterwalder and Pigneur, 2009).

In doing so, it bridges a gap between qualitative sufficiency frameworks and outcome-oriented environmental assessments. Qualitative models such as the "business for sufficiency framework" (Niessen and Bocken, 2021), empirical studies of sufficiency practices (Beyeler and Jaeger-Erben, 2022) and the "sufficiency-based circular economy" perspective (Bocken et al., 2022) provide rich conceptual guidance but do not offer scoring rules or thresholds for structured self-assessment. The scoring model introduced here complements these contributions by offering a pragmatic operationalization based on a limited number of parameters and a simple scoring logic. To date, no established industry-standard benchmarks or external reference values exist against which sufficiency maturity scores at the business-model level could be compared. Existing sustainability maturity frameworks, such as ESG-based assessment tools or sustainability indices, operate at the level of environmental and social outcomes and do not provide scoring rules specifically designed to assess the structural orientation of business models toward sufficiency. This absence of external benchmarks is not a peripheral limitation but reflects the broader measurement gap that motivates the present study. Within the scoring model, low sufficiency maturity is characterized by the absence or non-strategic implementation of sufficiency-relevant practices across parameters, reflected in scores of 0 on the majority of assessed dimensions. High sufficiency maturity, by contrast, indicates that sufficiency principles are structurally embedded across all three business-model dimensions, with practices systematically anchored in value proposition, value creation and delivery, and value capture, reflected in scores of 2 on the majority of parameters. As the empirical basis of the instrument broadens in future research, these qualitative characterisations can be progressively refined into sector-specific reference profiles.

Furthermore quantitative methods could be an alternative in measuring sufficiency. This could include comparison of the state of a product or process before and after implementing sufficiency measures using measurable indicators such as carbon footprint, material use, or energy demand. Life cycle assessment and before–after comparisons allow calculation of reductions, for example, lowered emissions per unit. These approaches provide comparability, transparency, support evidence-based decisions, and enable integration into sustainability reporting.

At the same time, the model does not claim to measure sufficiency in an absolute sense. Rather, it captures the implementation and strategic anchoring of sufficiency-relevant measures at business-model level. This focus on structural orientation aligns with calls to integrate sufficiency into corporate strategy and governance rather than treating it merely as a product-level design principle (Bocken et al., 2022; Gossen and Niessen, 2024).

5.2 Complementarity to LCA, ESG, and TBL-based approaches

The scoring model does not replace established assessment frameworks such as life-cycle assessment (LCA), absolute environmental sustainability assessments, ESG metrics or Triple Bottom Line (TBL)-oriented reporting. Rather, it addresses a different question by assessing the structural, business-model-level orientation toward sufficiency; parameter ratings (0–2) are aggregated into dimension scores (Value Proposition, Value Creation & Delivery, Value Capture), without computing an overall single score, while LCA evaluates outcome-level impacts along product life cycles and ESG/TBL integrate environmental, social and economic indicators at company or portfolio level (Bjørn et al., 2020; Crowther and Seifi, 2021; Diez-Cañamero et al., 2020; Niessen and Bocken, 2021; Richardson et al., 2023).

The model was developed as a tool for self-assessment of the sufficiency performance of manufacturing companies. Dimension scores help identify and communicate sufficiency strengths (e.g., long product lifetimes, repair and reuse services) and prioritize gaps (e.g., supply-chain sufficiency, sufficiency-oriented value capture). Used alongside established sustainability assessment and reporting approaches (e.g., LCA and ESG/TBL), the results support interpretation of whether observed improvements are underpinned by sufficiency-oriented changes in value logic or mainly reflect incremental efficiency and circularity gains within otherwise growth-oriented business models (Bocken et al., 2022; Gossen and Niessen, 2024; Mettler, 2011; Schmidhäuser et al., 2024).

At the same time, because the model captures implementation and strategic anchoring rather than absolute impacts, high dimension scores cannot be equated with alignment to planetary boundaries or sector-specific reduction pathways; a "score inflation" risk remains if firms still operate above acceptable absolute impact levels (Bjørn et al., 2020; Clément et al., 2022; OECD, 2025). If the model is used in isolation, it does not ensure that, even in the case of a high score, planetary boundaries are not exceeded. Future iterations of the model could therefore be linked to absolute environmental impact metrics to complement the structural sufficiency orientation captured by the scoring model with outcome-level environmental performance data. Accordingly, the scoring results should be embedded in broader sustainability management and reporting systems that incorporate absolute reference frames and rebound considerations, which are addressed here only qualitatively through open questions and conceptual framing (Bjørn et al., 2020; Wüst et al., 2022; Schmidhäuser et al., 2024; Tănase et al., 2025).

5.3 Practical implications for manufacturing firms

The scoring model can also support the integration of sufficiency into internal management processes. By aligning parameters with the business-model triad, the instrument can be linked to existing governance structures and responsibilities, for instance positioning need-oriented value propositions and non-manipulative communication within marketing and product management, and assigning sufficiency-oriented logistics and supplier criteria to operations and procurement (Niessen and Bocken, 2021; Gossen and Niessen, 2024). Over time, firms can track progress on sufficiency dimensions and link scores to internal KPIs, dashboards and decision-making processes. Beyond its function as a self-assessment instrument, the scoring model can be understood as a governance-enhancing artefact that supports the integration of sufficiency into organizational routines, decision-making structures, and accountability mechanisms. This alignment creates transparency about which organizational units carry responsibility for sufficiency-relevant practices and where governance gaps exist. Potential adoption barriers include limited internal awareness of sufficiency as a strategic concept, the absence of dedicated ownership for sufficiency-related parameters, and competing priorities in existing sustainability management systems. Conversely, adoption is facilitated when sufficiency parameters can be linked to existing KPI frameworks, when senior management endorses sufficiency as a strategic priority, and when the instrument is embedded in broader sustainability governance processes rather than applied as a standalone tool. The transparent scoring logic, in which each parameter rating is based on defined criteria and clearly defined scale levels, supports the traceability of assessment decisions and facilitates their integration into internal reporting and audit processes. The three-dimensional sufficiency profile lends itself to stakeholder-specific communication, for instance through radar diagrams, though the most appropriate format will depend on firm-specific communication practices and governance structures.

At the same time, the validation interviews showed that respondents tended to rate the sufficiency performance of their own companies rather optimistically. This tendency toward overestimation should not be interpreted as a deficiency of the respondents, but as a structural feature of self-assessment approaches. Consequently, this pattern of overestimation is a methodological limitation and potential weakness of the methodology.

5.4 Limitations and future research

Several limitations of the present study point to avenues for future work. First, the empirical validation is based on a purposive sample of five firms in the German manufacturing context. Consistent with DSR methodology, this sample size is appropriate for an initial design-evaluation cycle focused on artefact coherence and practical relevance rather than statistical generalizability (Venable et al., 2016; Peffers et al., 2007). Regarding inter-rater reliability, the self-assessment design, in which each firm evaluates its own business model, limits the direct application of classical inter-rater analyses. Future iterations could address this through cross-functional scoring workshops or external moderation to reduce interpretation variance.

Second, the self-assessment nature of the instrument entails methodological limitations. Scores rely on internal perceptions and may be influenced by optimism bias or differences in interpretation. The thesis therefore emphasizes that self-assessments should, where possible, be triangulated with objective indicators and documents. For future applications, combining individual self-assessments with cross-functional scoring workshops, external moderation or complementary external indicators could help to mitigate overestimation effects and strengthen the robustness of the results. To further reduce optimism bias, future applications could incorporate a structured triangulation approach in which self-assessment scores are substantiated with supporting documents such as revenue breakdowns, sustainability reports, or internal KPI dashboards, while preserving the lean character of the instrument.

Third, some parameters, particularly "sufficiency in supply chain and logistics" and "sufficiency-oriented value capture," require further operationalization. The validation highlights the need for more detailed guidance on measurement approaches, sector-specific benchmarks and evidence types. Future research could draw on empirical case studies and quantitative parameters to refine these aspects. Indicative thresholds may require sector-specific calibration, as the relevance and measurability of sufficiency parameters are likely to vary across industrial contexts. The three-point structure reflects a deliberate balance between discriminating power and definitional clarity: finer-grained scales would require more precise boundaries between adjacent levels than can currently be justified without broader empirical evidence and sector-specific benchmarks, and are therefore better suited to subsequent iterations of the model (Mettler, 2011).

Fourth, the current model does not integrate rebound effects or absolute environmental performance into the scoring logic. While the conceptual framing emphasizes the importance of rebound dynamics and planetary boundaries, these aspects are only captured qualitatively. A promising direction for future work is to link sufficiency scores to absolute environmental metrics or rebound-sensitive indicators to better reflect the interplay between structural orientation and actual impact (Lange et al., 2021; Richardson et al., 2023; Santarius, 2014).

Finally, the model is tailored to manufacturing firms and may require adaptation for other sectors, such as services or platform-based business models. Comparative studies could explore how sufficiency manifests differently across sectors and how the assessment logic needs to be modified accordingly. Further research could also explore how digital tools and automated data collection might support and streamline the self-assessment process in future iterations of the model. Future research should pursue a phased broadening of the empirical basis, extending validation first across additional manufacturing sectors and subsequently to cross-country contexts. Direct comparability of scores across sectors and countries should not be assumed *a priori*, as differences in industrial structures and regulatory environments are likely to necessitate sector-specific adaptations of parameter definitions and thresholds.

6 Conclusion

This paper has developed and empirically validated a business-model-level scoring model for industrial sufficiency in manufacturing firms. Grounded in sufficiency literature and business-model frameworks, the model translates sufficiency principles into eight assessment parameters across the dimensions of Value Proposition, Value Creation & Delivery, and Value Capture, operationalized on a three-point ordinal scale. Using a mixed-methods validation design with five manufacturing firms, the study demonstrates that the model is comprehensible and relevant for practitioners, can be applied as a lean self-assessment tool, and discriminates between different sufficiency maturity levels across parameters and dimensions.

The scoring model addresses a gap between qualitative sufficiency frameworks and outcome-oriented assessment approaches such as LCA, ESG metrics and TBL-based reporting. It provides a structural lens on how strongly manufacturing business models are oriented toward sufficiency, without replacing the need for absolute environmental performance assessments linked to planetary boundaries. For practitioners, the model offers a starting point for integrating sufficiency into strategic steering and internal communication. For researchers, it represents an operational artefact that can be further tested, refined and combined with other assessment tools.

Future research should broaden the empirical basis, refine operationalization for challenging parameters, explore links to absolute environmental metrics and rebound-sensitive indicators, and adapt the model for other sectors. In doing so, it can contribute to making sufficiency not only a normative ambition but a measurable and manageable component of sustainable business models in industry.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Author contributions

RP: Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing. PS: Conceptualization, Methodology, Supervision, Validation, Visualization, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that Generative AI was used in the creation of this manuscript. The author declares the use of generative artificial intelligence in the preparation of this manuscript. Specifically, ChatGPT (OpenAI; large language model; model/version: GPT-5.2; source: ChatGPT web interface; accessed on: [25.11.25 – 23.12.25]) and Microsoft 365 Copilot, based on ChatGPT 5 was used to support manuscript drafting by summarizing, shortening, and paraphrasing selected sections, translation of German sections to English and by improving wording and readability. The tool was not used to generate or analyze empirical data, to produce figures or tables. All AI-assisted outputs were critically reviewed, verified, and edited by the author, who takes full responsibility for the accuracy, originality, and integrity of the final manuscript, including citations and references.

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